

TANVI MEHETRE

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SUMMARY

Software Engineer with 1 year of experience in cloud-native and full-stack development, specializing in microservices. Experienced in building event-driven architectures using AWS EventBridge and developing enterprise web applications with ASP.NET MVC, C#, and JavaScript. Strong foundation in algorithms, frontend and backend engineering, with hands-on experience in performance optimization, API development, and cross-functional collaboration.

TECHNICAL SKILLS

EDA & Hardware Tools: Genus (Synthesis), Cadence Innovus (PnR), Questa (Simulation & Verification), Cadence Virtuoso, LTSPICE

HDL & RTL Design: Verilog, VHDL, RTL design, Synthesis, Place-and-route, Timing analysis, Power optimization

Programming Languages: Python, Java, C#, JavaScript, C, SQL

Systems & OS: Linux, Windows, MacOS

Distributed & Systems: RPC, XDR, CORBA, Multithreaded / Distributed Programming, Microprocessor Architecture

Software & Cloud: ASP.NET MVC, FastAPI, React.js, Node.js, Next.js, AWS, PostgreSQL, MongoDB, Git, Databricks, DAX, Power BI

Computer Science Fundamentals: Data Structures, Algorithms, Software Development Lifecycle

EDUCATION

Iowa State University, Ames, Iowa, USA
Bachelor of Science, Computer Engineering

GPA: 3.55/4
Graduation: May 2025

PROFESSIONAL EXPERIENCE

Software Engineer – Quadyster, Bettendorf, Iowa, USA

August 2025 – Present

AvMC Power BI Workshop – Instructor

- Co-instructed a Power BI workshop at Redstone Arsenal, Huntsville, Alabama, USA as part of an instructor team.
- Delivered sessions covering the Power BI interface, data transformation, modeling, and visualizations, facilitated hands-on exercises to reinforce participant learning.

Enterprise Connected Data Domain (John Deere)

- Led key tasks across the software development lifecycle, designing a scalable, event-driven system architecture using AWS EventBridge to support enterprise order management workflows across microservices.
- Enhanced event schemas, routing logic, and service interactions using modern AI-assisted development practices to support reliable communication, deployed backend infrastructure components in a cloud-native environment.

Project Time (eTime) Application

- Designed and implemented backend REST APIs for a serverless employee project time-tracking system using *FastAPI* and *AWS Lambda*, with *PostgreSQL* for persistent data storage.
- Developed responsive frontend in *React.js* with Redux state management, validation logic, and thorough testing using React Testing Library.

Fullstack Programmer Intern – Iowa Department of Transportation, Ames, Iowa, USA

April 2024 – May 2025

- Developed and maintained LIMS and MAPLE enterprise projects using *ASP.NET MVC*, *C#*, *SQL*, and *JavaScript*, contributing to integration across large-scale software systems and backend services.
- Applied problem-solving aptitude to deliver 7 bug fixes, 15 feature enhancements, and 10 code refactors in three months; participated in code reviews, bug triage, and production support for backend applications.

Student System Specialist – Iowa State University, Ames, Iowa, USA

September 2022 – August 2024

- Provided technical support for COM S and WLC departments, diagnosing and resolving hardware and software issues at the system level on *Linux* and Windows platforms.

- Assisted with system setups, installations, and network troubleshooting in cross-functional team environments.

PROJECTS

Digital Design & Physical Design for Real-Time Statistical Analysis

Developed a Verilog-based RTL system to compute moving average and standard deviation of temperature samples.

- Synthesized design in Genus, performed Place-and-Route in Cadence Innovus, and simulated in Questa directly mirroring the Genus Synthesis Solution toolchain.
- Optimized design across iterations: achieved 20% area reduction and 50% power consumption reduction through iterative synthesis and timing closure.
- Generated control signals for sample readiness and output completion, demonstrating design flow proficiency.

Microprocessor Development (Single-Cycle and Multi-Cycle Architectures)

Designed and implemented a full microprocessor in VHDL using both single-cycle and multi-cycle architectures.

- Designed and implemented a full microprocessor in VHDL using both single-cycle and multi-cycle architectures.
- Performed functional simulation and timing verification in Questa, collaborating with peers to validate processor microarchitecture and ensure correct instruction execution.
- Optimized operating frequency from 50 MHz (single-cycle) to 1 GHz (multi-cycle), showcasing architecture-level performance tuning relevant to static timing analysis.

The Spinning Cyclone (Net Deployment System)

Contributing to the COSMIC Capstone Challenge (C3) team for in-space servicing and manufacturing (ISAM) payload.

- Designed a system with prototyping in CAD utilizing spring-loaded masses and cold gas thrusters to address the space debris issue.
- Developed a scalable solution for capturing and deorbiting space debris.

Remote System Monitoring via RPC

Built a distributed client-server system in C for real-time remote machine monitoring using RPC.

- Implemented custom RPC interfaces and used *XDR* for structured data exchange between *Linux* machines, demonstrating multithreaded and distributed programming skills.
- Retrieved and displayed real-time CPU, memory and load information with accurate remote querying.

LLM-Based File System Bug Classification Report

Researched and tried to build a system to automate bug classification using Large Language Models to improve large-scale debugging workflows.

- Fine-tuned LLMs on historical bug reports (e.g., PatchDB) to predict and categorize file system issues at scale.
- Benchmarked model outputs against expert analysis to quantify accuracy and identify improvements in automated debugging pipelines.

Pursue-Escape Grid Simulation (Greedy and Minimax)

Developed real-time adversarial search agents in Python using heuristic-based Greedy and Minimax algorithms.

- Implemented Greedy and Minimax search algorithms in Python for a multi-agent grid-based simulation with depth-limited decision logic.
- Designed heuristic evaluation functions demonstrating strong analytical and algorithmic problem-solving skills.

Distributed Order Management System (CORBA)

Designed a client-server system in Java using CORBA for distributed order placement, tracking, and management.

- Implemented manager and customer roles with real-time order visibility and updates across distributed nodes.
- Handled error propagation, role separation, and distributed state consistency in a multi-client environment.

Autonomous Robot (Tree Cutting Simulation)

Programmed real-time embedded control logic for an autonomous robot using UART-based sensor integration.

- Implemented sensor-driven decision logic (cliff, width, and distance detection) directly on hardware using *C*.
- Successfully tested in a real-world environment, demonstrating hardware-software co-development expertise.

Globe Gatherer (Android Travel Application)

Contributed to a team-based Android application developed in Java.

- Built user-facing features for travel planning using Java and Android Studio.

- Implemented admin functionality for user management and announcements.

ACTIVITIES

Engineering Student Council

May 2023 – May 2025

Vice President of Administration

- Represented Iowa State University at the NAESC conference in New Mexico and Minnesota.
- Led administrative coordination across engineering department clubs.

Digital Women Club

Dec 2024 – May 2025

Secretary

- Won Best Overall Project at Digital Women Code Camp